

Optimization exam
May 2026

Exercise 1. 1 We define a circumcenter of a set of points $x_0, x_1, \dots, x_m \in \mathbb{R}^n$ as a point which is equidistant to $x_0, x_1, \dots, x_m$, and which belongs to the affine space generated by these points. Show that if $x_0, x_1, \dots, x_m$ are affine independent then the circumcenter is uniquely defined.

Exercise 2. The objective of this exercise is to prove the convergence of the following algorithm.

Algorithm 1

1: **Input:** strongly convex function f , $x_1 \in \mathbb{R}^n$, $\varepsilon > 0$; set $k = 1$.

2: **Step 1.** If $\|\nabla f(x_k)\| \leq \varepsilon$ stop and return x_k .

3: **Step 2.** Find $t_k > 0$ via bisection with

$$f(x_k - t_k \nabla f(x_k)) = f(x_k).$$

Set

$$y_k = x_k - t_k \nabla f(x_k).$$

4: **Step 3.** If $\nabla f(x_k)$ and $\nabla f(y_k)$ are linearly independent, solve

$$(\alpha_k, \beta_k) = \operatorname{argmin}_{(\alpha, \beta) \in \mathbb{R}^2} f(x_k + \alpha \nabla f(x_k) + \beta \nabla f(y_k))$$

and set

$$x_{k+1} = x_k + \alpha_k \nabla f(x_k) + \beta_k \nabla f(y_k).$$

5: **Step 3.** If $\nabla f(x_k)$ and $\nabla f(y_k)$ are linearly dependent, set x_{k+1} to either

(a) the minimizer of

$$f(x_k - \lambda t_k \nabla f(x_k))$$

over $\lambda \in [0, 1]$ or

(b) the midpoint

$$x_k - \frac{t_k}{2} \nabla f(x_k).$$

6: **Step 4.** Set $k \leftarrow k + 1$ and go to Step 1.

We will assume that f is L -smooth, i.e., there is $L > 0$ such that

$$\|\nabla f(y) - \nabla f(x)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

We will also admit the following inequality (Baillon-Haddad) which holds for a μ -strongly convex and L -smooth function:

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle + \frac{\|\nabla f(y) - \nabla f(x)\|^2}{2L},$$

for all $x, y \in \mathbb{R}^n$.

Definition 0.1 (Line) . A line in $\mathbb{R}^n$ is $\{tp + (1 - t)q : t \in \mathbb{R}\}$ for $p \neq q$ in $\mathbb{R}^n$.

1) Let f be a L -smooth and μ -strongly convex function with minimizer x^* . Show that

1.1) 0.5 $f(x - \frac{1}{L}\nabla f(x)) \leq f(x) - \frac{\|\nabla f(x)\|^2}{2L};$

1.2) 0.75 (Polyak-Lojasiewicz) $\|\nabla f(x)\|^2 \geq 2\mu(f(x) - f(x^*));$

1.3) 0.5 $f(x) - f(x^*) \geq \frac{\mu}{2}\|x - x^*\|^2;$

1.4) 0.5 $f(x) - f(x^*) \leq \frac{L}{2}\|x - x^*\|^2;$

1.5) 0.5 $\|\nabla f(x)\|^2 \leq 2L(f(x) - f(x^*)).$

2) 0.75 Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be μ -strongly convex. For any $c \in \mathbb{R}$ and any line $\mathcal{L}$, show that the set $\mathcal{L} \cap \{f = c\}$ has at most two elements.

3) 1.25 Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable and μ -strongly convex. Let $x \in \mathbb{R}^n$ with $\nabla f(x) \neq 0$. Show that there exists a unique $t > 0$ such that $f(x - t\nabla f(x)) = f(x)$.

4) Suppose $\nabla f(x_k)$ and $\nabla f(y_k)$ are linearly independent.

4.1) 0.25 Show that

$$\langle \nabla f(x_{k+1}), \nabla f(x_k) \rangle = 0 \quad \text{and} \quad \langle \nabla f(x_{k+1}), \nabla f(y_k) \rangle = 0.$$

4.2) 0.5 Show that

$$\|\nabla f(x_{k+1}) - \nabla f(x_k)\|^2 = \|\nabla f(x_{k+1})\|^2 + \|\nabla f(x_k)\|^2 \leq L^2\|x_{k+1} - x_k\|^2.$$

4.3) 0.5 Show that

$$f(x_k) - f(x_{k+1}) \geq \frac{\|\nabla f(x_{k+1})\|^2 + \|\nabla f(x_k)\|^2}{2L}.$$

5) Let f be L -smooth and μ -strongly convex with $0 < \mu < L$, and let $\{x_k\}$ be the sequence generated by the algorithm. Denote

$$\eta = 1 - \frac{\mu}{L}, \quad \kappa = \frac{L}{\mu}, \quad \text{and} \quad \eta^* = \frac{\kappa - 1}{\kappa + 1}.$$

5.1) 1 Show that

$$f(x_k) - f(x^*) \leq \eta^{k-1}(f(x_1) - f(x^*)) \quad \text{for all } k \geq 1.$$

5.2) 1 Show that at any step k where $\nabla f(x_k)$ and $\nabla f(y_k)$ are linearly independent,

$$f(x_{k+1}) - f(x^*) \leq \frac{\kappa - 1}{\kappa + 1}(f(x_k) - f(x^*)) = \eta^*(f(x_k) - f(x^*)).$$

5.3) 1 Show that

$$\|x_k - x^*\|^2 \leq \frac{L}{\mu} \eta^{k-1} \|x_1 - x^*\|^2.$$